

# Model Evaluation Metrics

Evaluating the performance of a machine learning model is a critical step that determines whether the model is ready for deployment or requires further refinement. For classification tasks, accuracy, which measures the proportion of correct predictions out of all predictions made, is one of the simplest and most commonly used metrics, though it can be misleading when working with imbalanced datasets where one class significantly outnumbers others. Precision measures the proportion of positive predictions that were actually correct, making it particularly important in situations where false positives carry a high cost. Recall, also known as sensitivity, measures the proportion of actual positive cases that were correctly identified by the model, which becomes especially important in situations where failing to identify a positive case, such as a disease diagnosis, carries serious consequences. The F1 score combines precision and recall into a single metric by calculating their harmonic mean, providing a balanced measure that is particularly useful when there is an uneven class distribution within the dataset. A confusion matrix provides a detailed breakdown of a classification model's performance, showing the counts of true positives, true negatives, false positives, and false negatives, offering deeper insight than a single summary metric alone. For regression tasks, common evaluation metrics include mean absolute error, which measures the average magnitude of prediction errors, and root mean squared error, which penalizes larger errors more heavily due to the squaring of differences before averaging. Cross-validation, a technique where a dataset is split into multiple subsets and a model is trained and evaluated multiple times using different combinations of these subsets, helps provide a more reliable estimate of how a model is likely to perform on new, unseen data compared to a single train-test split.